choosing filters

data prep

the dataset is loaded and curred. We remove adult = 0 and NAs

```
data250_afabae_00 <- read.csv("../../../../01.data/2023_screen_specifics_251genotypes/02.aphi
  mutate(ivip = as.factor(ivip)) %>%
  mutate(session = as.factor(session))



data250_afabae_01 <- data250_afabae_00 %>% filter(!is.na(larvae_total_number))
data250_afabae_01$ivip <- droplevels(data250_afabae_01$ivip)

data250_afabae_02<- data250_afabae_01 %>% filter(adult_alive != "0") #remove if no adult
data250_afabae_02$ivip <- droplevels(data250_afabae_02$ivip)

data250_afabae_02$ivip <-as.factor(data250_afabae_02$ivip)
```

Once we have removed the NAs and all rows for which we don't have an adult, we can still make adjustments to the dataset. We can remove outliers to ensure that our distribution approaches a normal distribution, and we can also filter out genotypes based on a threshold for the minimum number of repetitions required to keep a genotype.

The goal of this script is to study how our model reacts to changes in the outlier removal rate and the minimum number of repetitions required to retain a genotype.

in the folowing loop we varry two parameters, i and x the coef of stringensy for the outliers removal and the minimal number of reps for the genotypes respectively.

when x = 1 we remove a lot of outlier and when x = 2 almost no outliers are removed

```r
x_values <- seq(1, 2, by = 0.05)

# Initialize an empty data frame to store results
results <- data.frame(x = numeric(0), i = numeric(0), AIC = numeric(0), BIC = numeric(0), RMS
                      n_before = numeric(0), n_after = numeric(0), n_diff = numeric(0), H2=nu

# Loop through different values of x
for (x in x_values) {
  for (i in 3:7) {
    # Calculate IQR and bounds for outlier removal
    IQR_value <- IQR(data250_afabae_02$larvae_total_number)
    lower_bound <- quantile(data250_afabae_02$larvae_total_number, 0.25) - x * IQR_value
    upper_bound <- quantile(data250_afabae_02$larvae_total_number, 0.75) + x * IQR_value

    # Filter out outliers
    data <- data250_afabae_02[data250_afabae_02$larvae_total_number >= lower_bound & data250_

    data <- data %>%
      group_by(ivip) %>%           # Group by genotype
      filter(n() >= i) %>%             # Keep only groups with i or more repetitions
      ungroup()


    data$ivip <- droplevels(data$ivip)

    NGEN<-nlevels(data$ivip)


    # Number of rows before removing outliers
    n_before <- nrow(data250_afabae_02)


    # Number of rows after removing outliers
    n_after <- nrow(data)

    # Calculate the difference in number of observations
    n_diff <- n_before - n_after

    # Fit the model
    mod <- lmer(larvae_total_number ~ ivip + (1|session/block), data = data)

    # Extract fitted values and residuals
```

```
    fitted_values <- fitted(mod)
    residuals_values <- residuals(mod)

    # Perform the Shapiro-Wilk test
    shapiro_result <- shapiro.test(residuals_values)

    # Print the p-value (this is your quantification of normality)
    shapiro <- shapiro_result$p.value

    # Calculate AIC and BIC
    model_AIC <- AIC(mod)
    model_BIC <- BIC(mod)

    # Calculate RMSE (Root Mean Squared Error)
    RMSE <- sqrt(mean(residuals_values^2))

    # Fit the model with a fixed formula
    mod_h2 <- lmer(larvae_total_number ~ (1 | ivip) + (1 | session), data = data)

    # Get the variance components
    estimates <- as.data.frame(VarCorr(mod_h2))

    # Extract the relevant variance components
    genetic_variance <- estimates$vcov[1]
    residual_variance <- estimates$vcov[3]  # Adjust based on the placement of residual varia

    nbE <-nlevels(data$session)

    # Calculate heritability (h2)
    H2 <- genetic_variance / (genetic_variance + (residual_variance / nbE))



    # Store the results in the data frame
    results <- rbind(results, data.frame(x = x, i = i, AIC = model_AIC, BIC = model_BIC, RMSE
                                         n_before = n_before, n_after = n_after, n_diff = n_d
  }


}
```

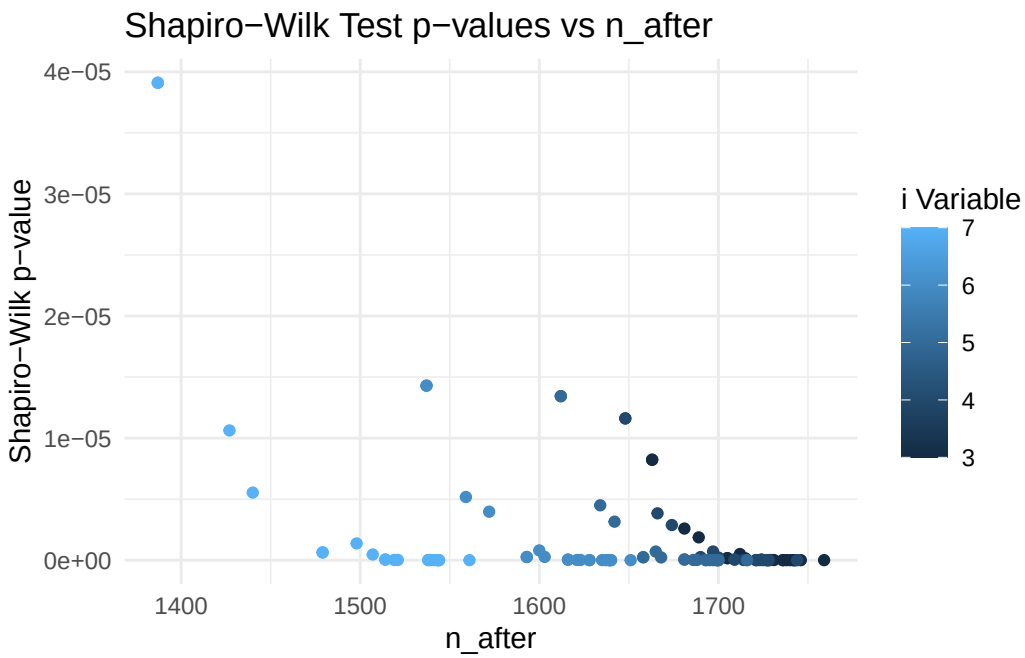
```
Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
```

```
Model failed to converge with max|grad| = 0.00233547 (tol = 0.002, component 1)
```

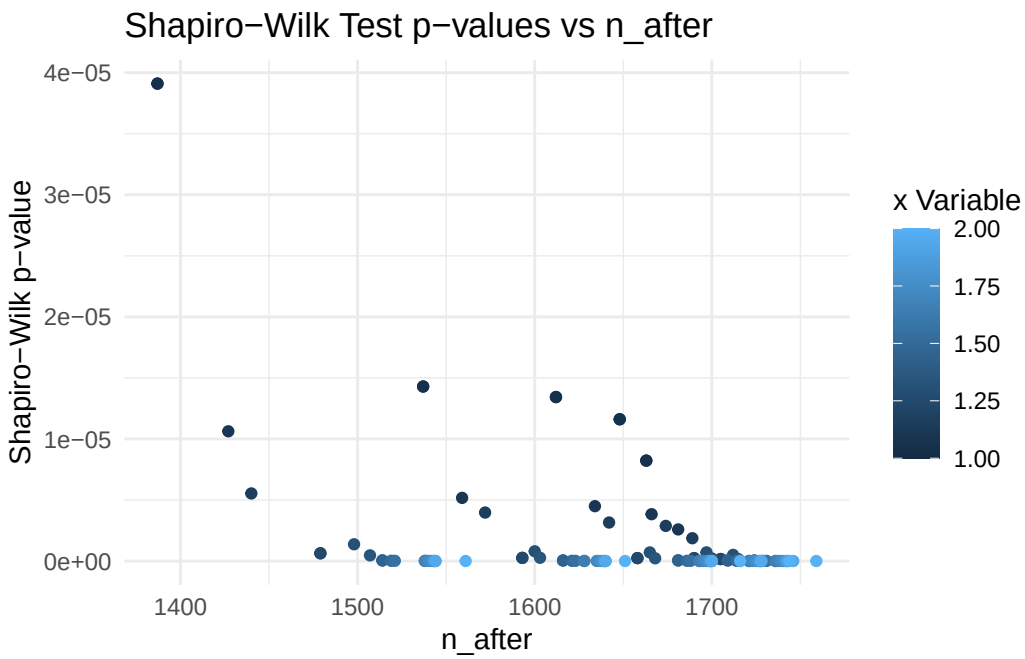
the file results contain all the data we need

to annalyse it we can plot different parameters also we can colorate the graph based on the value of i or x to understand the patern and impact of these 2 parameters.

```
ggplot(results, aes(x = n_after, y = shapiro, color = i)) +
  geom_point() +  # Scatter plot with colored points based on 'i'
  labs(title = "Shapiro-Wilk Test p-values vs n_after",
       x = "n_after",
       y = "Shapiro-Wilk p-value",
       color = "i Variable") +  # Color legend label
  theme_minimal()
```

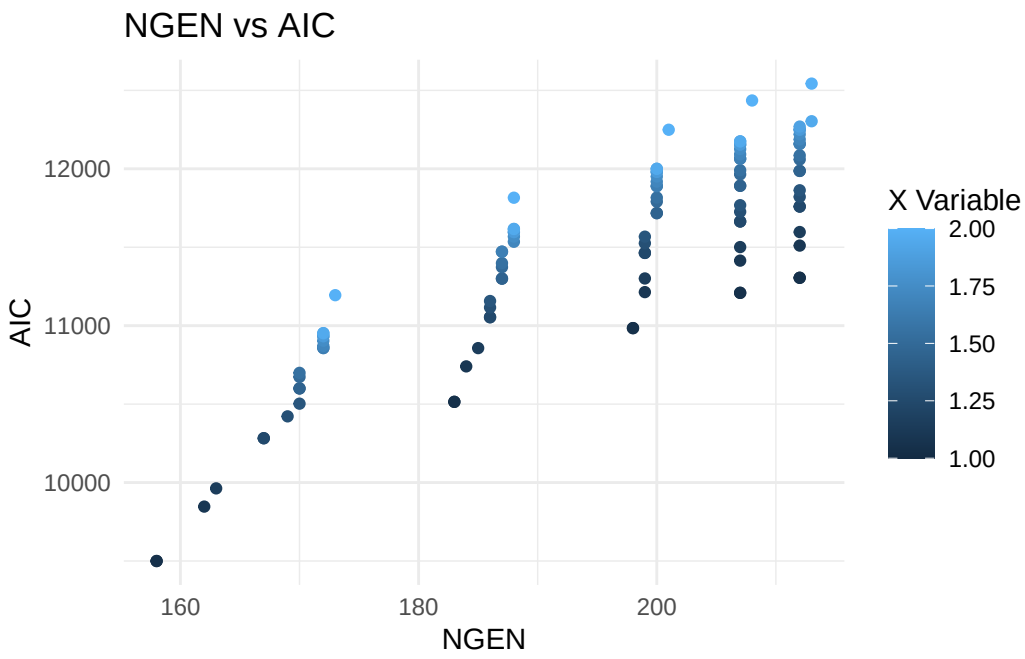



```
ggplot(results, aes(x = n_after, y = shapiro, color = x)) +
  geom_point() +  # Scatter plot with colored points based on 'x'
  labs(title = "Shapiro-Wilk Test p-values vs n_after",
       x = "n_after",
       y = "Shapiro-Wilk p-value",
       color = "x Variable") +  # Color legend label
  theme_minimal()
```

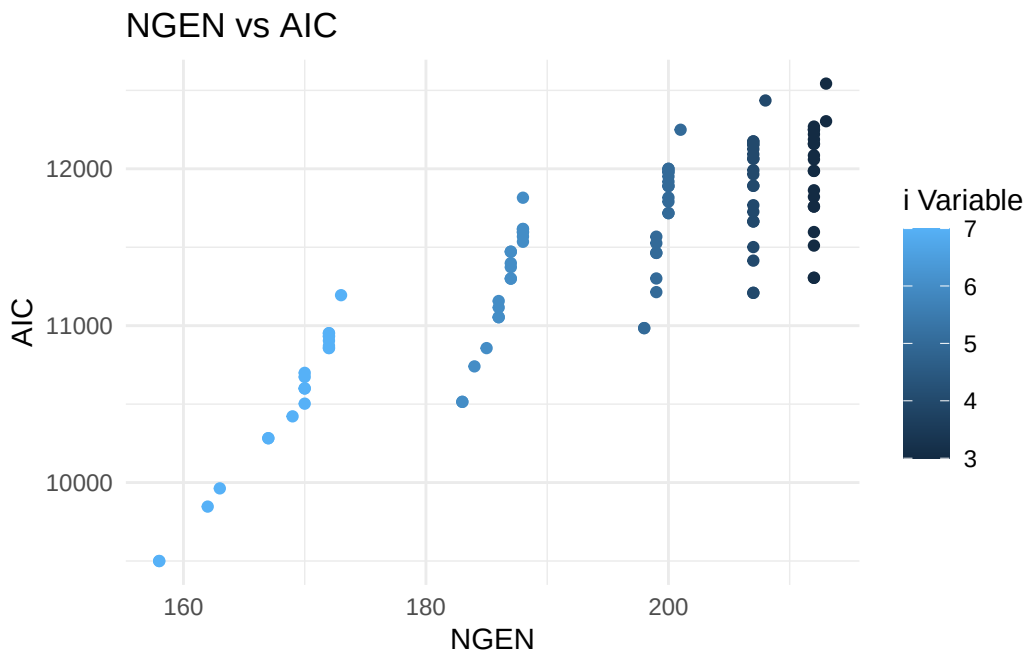


a high shapiro value indicate a more normal distribution of the residuals

```
ggplot(results, aes(x = NGEN, y = AIC, color = x)) +
  geom_point() +  # Scatter plot with colored points based on 'x'
  labs(title = "NGEN vs AIC",
       x = "NGEN",
       y = "AIC",
       color = "X Variable") +  # Color legend label
  theme_minimal()
```

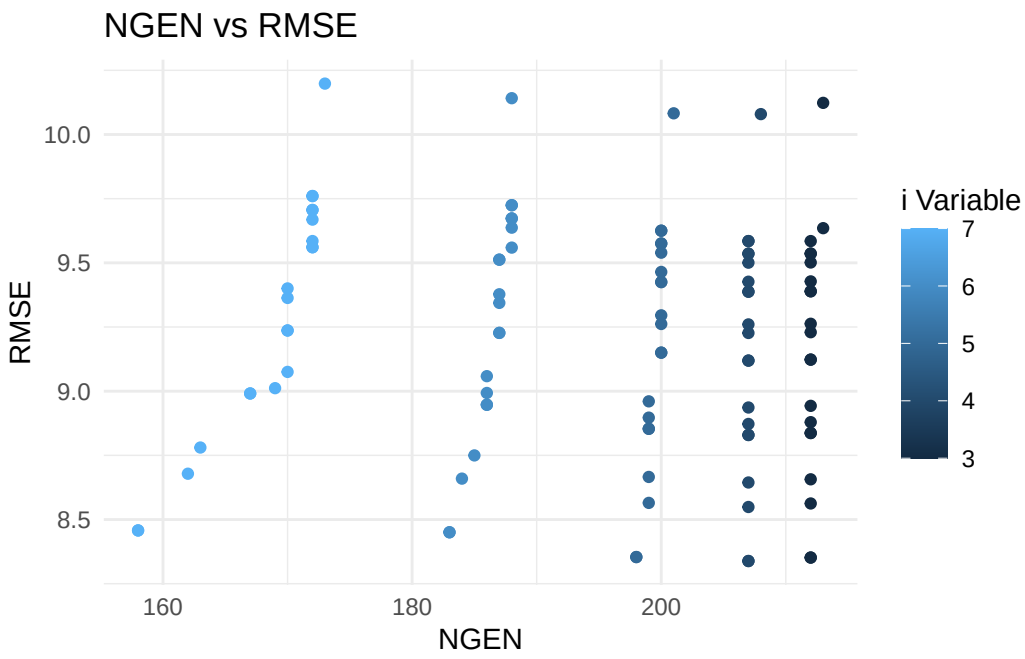



```
ggplot(results, aes(x = NGEN, y = AIC, color = i)) +
  geom_point() +  # Scatter plot with colored points based on 'i'
  labs(title = "NGEN vs AIC",
       x = "NGEN",
       y = "AIC",
       color = "i Variable") +  # Color legend label
  theme_minimal()
```

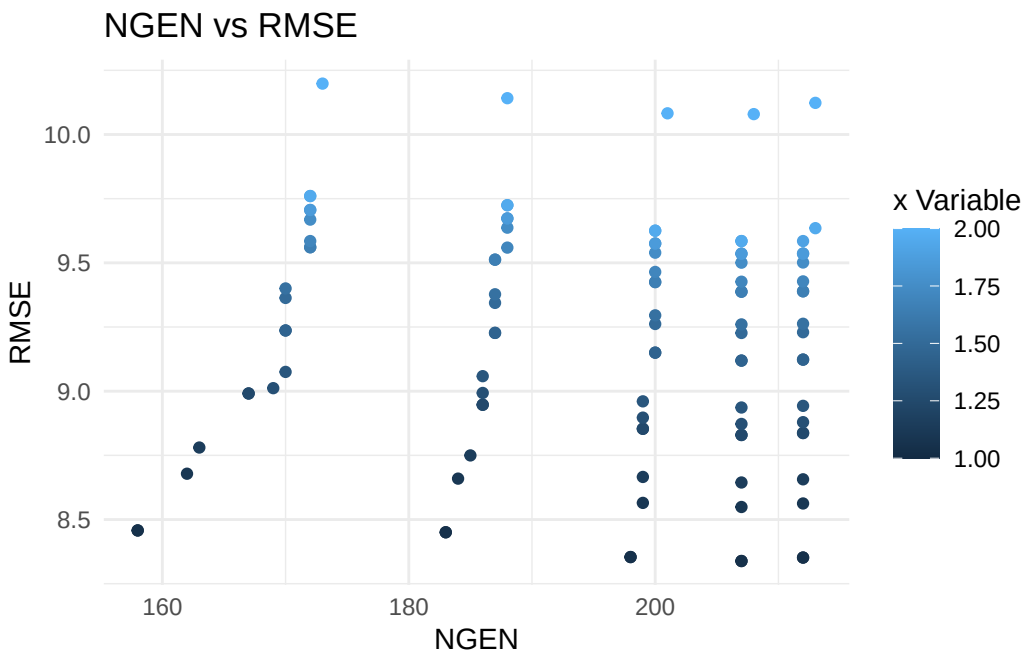


ngen = how many genotypes we keep

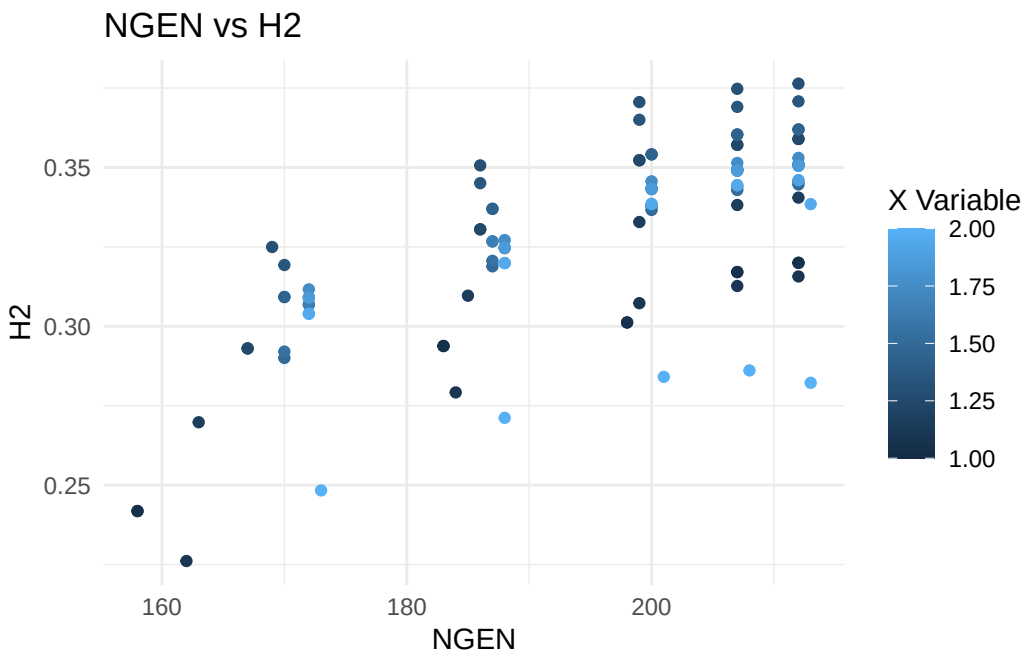
```
ggplot(results, aes(x = NGEN, y = RMSE, color = i)) +
  geom_point() +  # Scatter plot with colored points based on 'i'
  labs(title = "NGEN vs RMSE",
       x = "NGEN",
       y = "RMSE",
       color = "i Variable") +  # Color legend label
  theme_minimal()
```



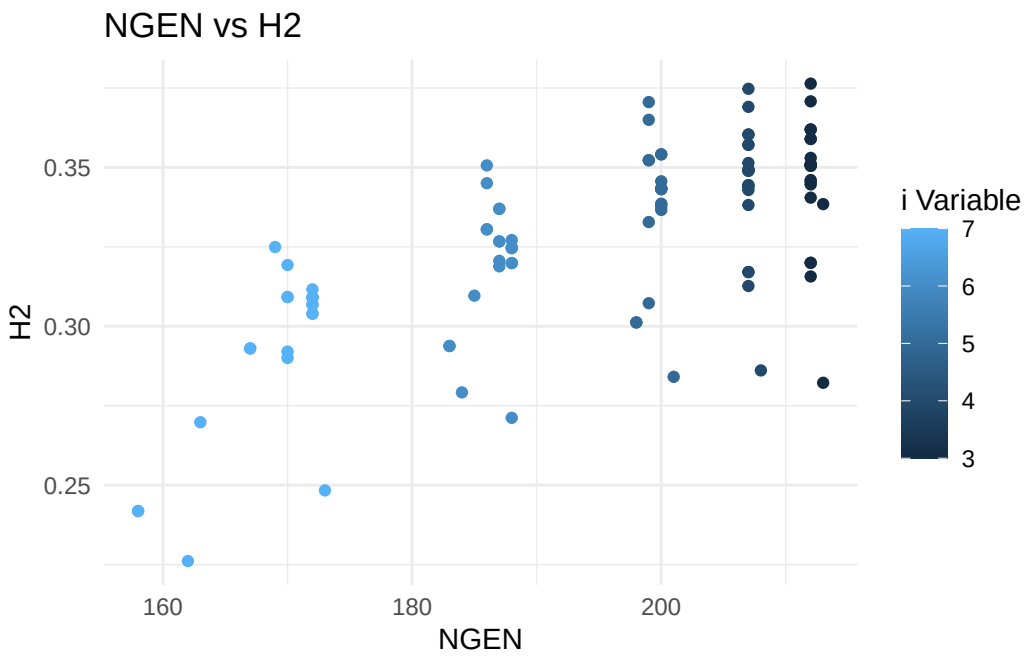

```
ggplot(results, aes(x = NGEN, y = RMSE, color = x)) +
  geom_point() +  # Scatter plot with colored points based on 'x'
  labs(title = "NGEN vs RMSE",
       x = "NGEN",
       y = "RMSE",
       color = "x Variable") +  # Color legend label
  theme_minimal()
```



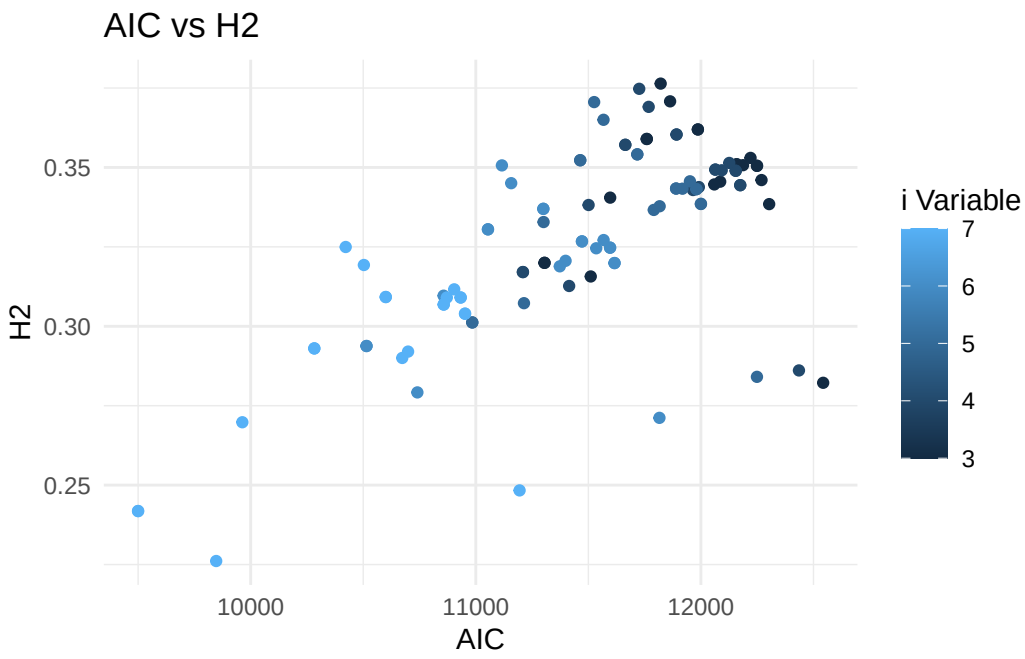

```
ggplot(results, aes(x = NGEN, y = H2, color = x)) +
  geom_point() +  # Scatter plot with colored points based on 'x'
  labs(title = "NGEN vs H2",
       x = "NGEN",
       y = "H2",
       color = "X Variable") +  # Color legend label
  theme_minimal()
```



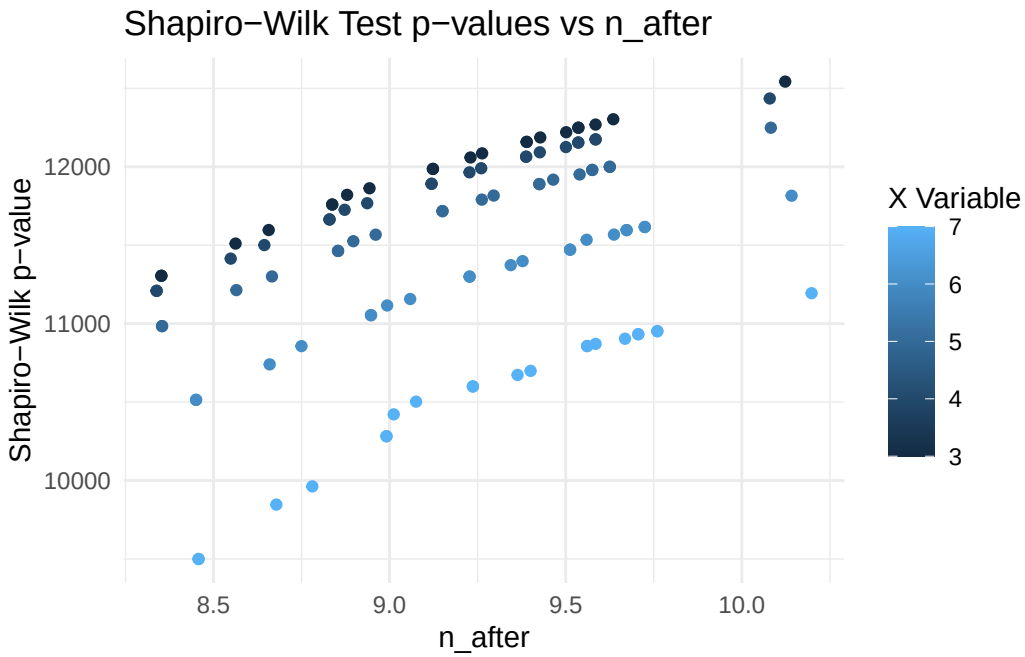

```
ggplot(results, aes(x = NGEN, y = H2, color = i)) +
  geom_point() +  # Scatter plot with colored points based on 'i'
  labs(title = "NGEN vs H2",
       x = "NGEN",
       y = "H2",
       color = "i Variable") +  # Color legend label
  theme_minimal()
```




```
ggplot(results, aes(x = AIC, y = H2, color = i)) +
  geom_point() +  # Scatter plot with colored points based on 'x'
  labs(title = "AIC vs H2",
       x = "AIC",
       y = "H2",
       color = "i Variable") +  # Color legend label
  theme_minimal()
```




```
ggplot(results, aes(x = RMSE, y = AIC, color = i)) +
  geom_point() +  # Scatter plot with colored points based on 'i'
  labs(title = "Shapiro-Wilk Test p-values vs n_after",
       x = "n_after",
       y = "Shapiro-Wilk p-value",
       color = "X Variable") +  # Color legend label
  theme_minimal()
```


Shapiro−Wilk Test p−values vs n_after
Shapiro−Wilk p−value
12000
11000
10000
8.5
9.0
9.5
10.0
n_after
X Variable
7
6
5
4
3